

Environmental Variables & Data Exploration

University of Florida

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UF | UNIVERSITY of FLORIDA

FLORIDA MUSEUM

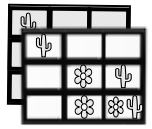
iDigBio Integrated Digitized Biocollections

NSF

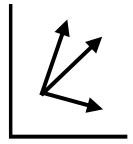
Agenda



Finding Climate/Abiotic Data Layers



Extracting Climatic Variables



PCA



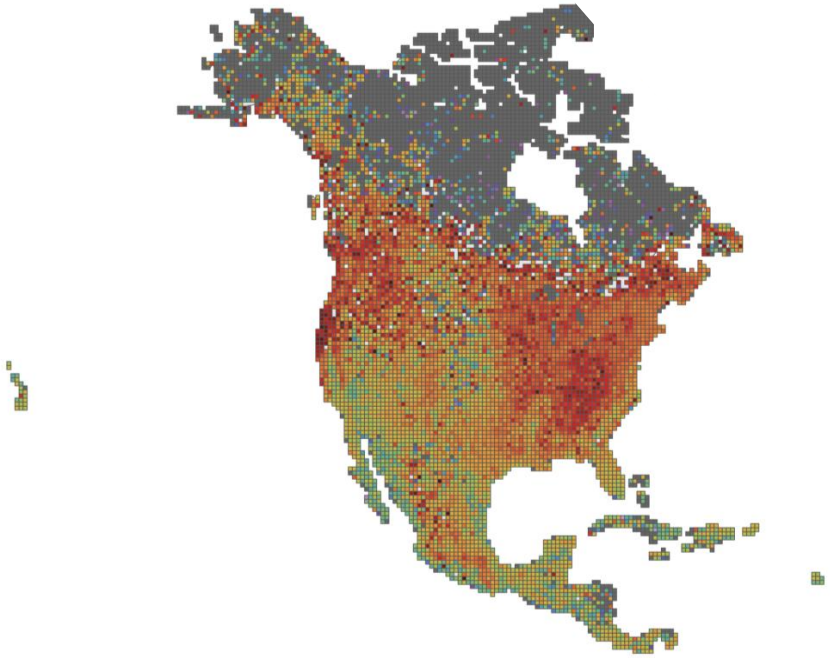
ANOVA



Coding Practical

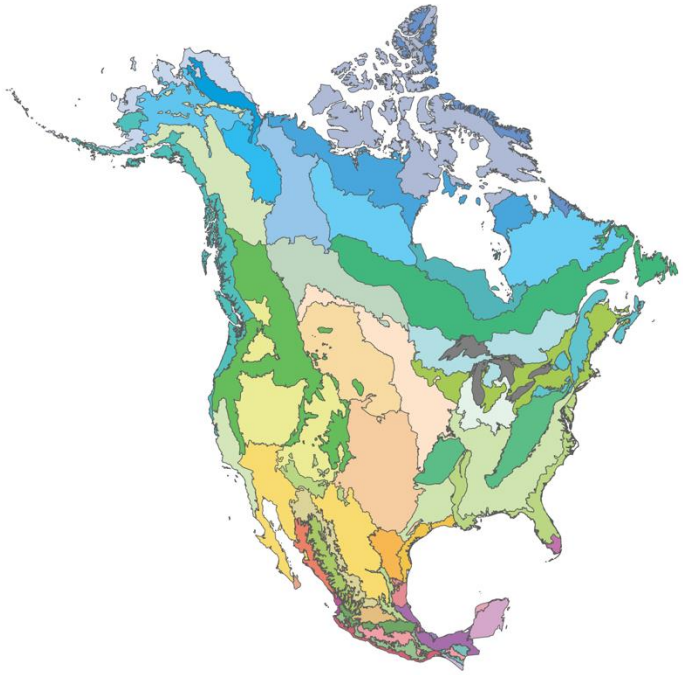
Climatic Variables/Abiotic Data Layers

Rasters



"grid data"

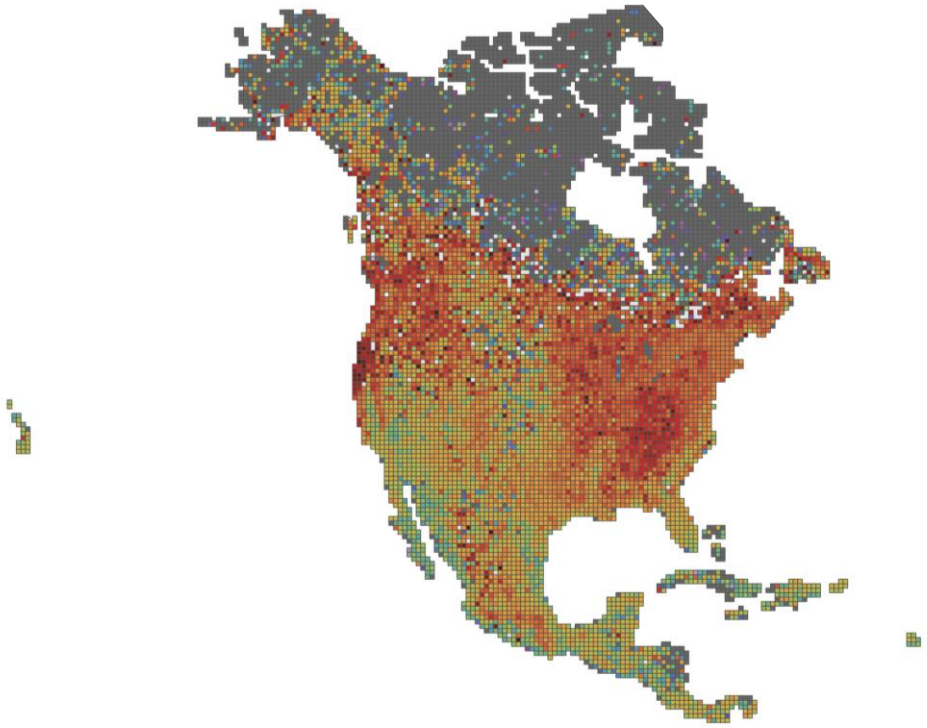
Vectors



"shape data"

Climatic Variables/Abiotic Data Layers

Rasters



“grid data”

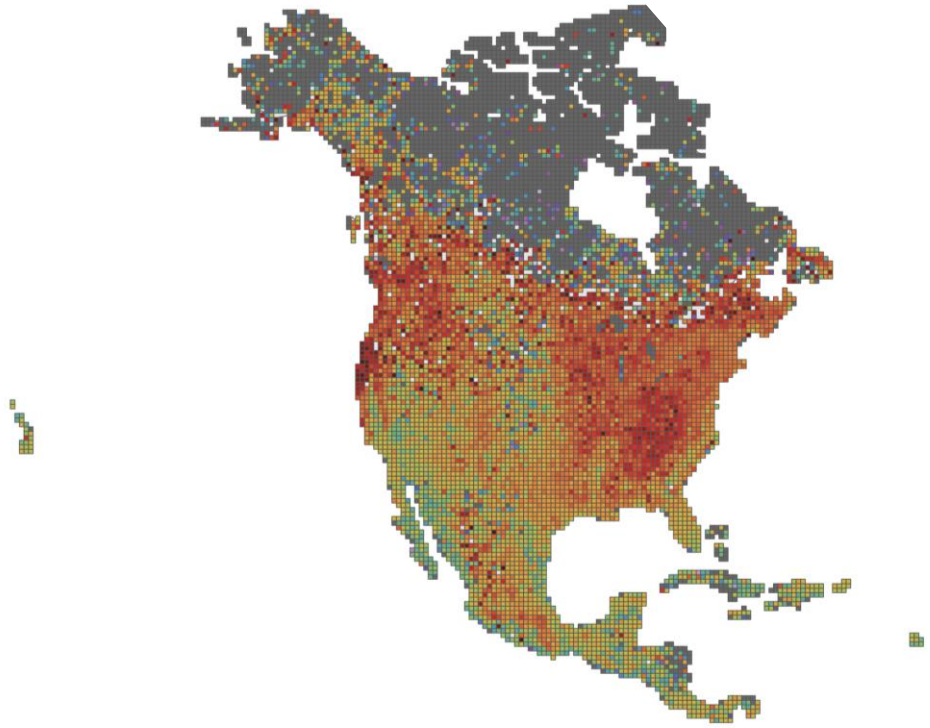
Vectors



“shape data”

Climatic Variables/Abiotic Data Layers

Rasters



"grid data"

Raster Resolution Note:

- Arc-seconds/minutes
  - Easy to work with latitude/longitude data
  - Not equal-area

30 seconds ≈ 1km^2

Climatic Variables/Abiotic Data Layers

What?

- Temperature
- Precipitation
- Soil
- Vegetation
- Land use

Where?

- Local
- USA
- North America
- Global

When?

- Past
- Current
- Future

Climatic/Abiotic Data Layer Sources

	What?	*Where?*	*When?*
• Worldclim2	*19 bioclimatic variables*	*Global*	*Past, Current, Future*
• PRISM	*Weather data*	*United States*	*Past, Current*
• Soil Grids	*Soil properties*	*Global*	*Current*
• Web Soil Survey	*Soil characteristics*	*United States*	*Current*

Data Layer Sources: WorldClim(2)

https://www.worldclim.org/data/bioclim.html

WorldClim

What? • 19 Bioclimatic variables
- Derived from monthly temperature & rainfall values
- Multiple resolution options!

Where? • Global land area
- Except Antarctica

When? • Past, Current, and Future projections

BIO1 = Annual Mean Temperature	BIO12 = Annual Precipitation
BIO2 = Mean Diurnal Range (Mean of monthly (max temp - min temp))	BIO13 = Precipitation of Wettest Month
BIO3 = Isothermality (BIO2/BIO7) (×100)	BIO14 = Precipitation of Driest Month
BIO4 = Temperature Seasonality (standard deviation ×100)	BIO15 = Precipitation Seasonality (Coefficient of Variation)
BIO5 = Max Temperature of Warmest Month	BIO16 = Precipitation of Wettest Quarter
BIO6 = Min Temperature of Coldest Month	BIO17 = Precipitation of Driest Quarter
BIO7 = Temperature Annual Range (BIO5-BIO6)	BIO18 = Precipitation of Warmest Quarter
BIO8 = Mean Temperature of Wettest Quarter	BIO19 = Precipitation of Coldest Quarter
BIO9 = Mean Temperature of Driest Quarter	
BIO10 = Mean Temperature of Warmest Quarter	
BIO11 = Mean Temperature of Coldest Quarter	

The data is available at the four spatial resolutions, between 30 seconds (~1 km2) to 10 minutes (~340 km2). Each download is a "zip" file containing 12 GeoTiff (.tif) files, one for each month of the year (January is 1; December is 12).

variable	10 minutes	5 minutes	2.5 minutes	30 seconds
minimum temperature (°C)	tmin 10m	tmin 5m	tmin 2.5m	tmin 30s
maximum temperature (°C)	tmax 10m	tmax 5m	tmax 2.5m	tmax 30s
average temperature (°C)	tavg 10m	tavg 5m	tavg 2.5m	tavg 30s
precipitation (mm)	prec 10m	prec 5m	prec 2.5m	prec 30s
solar radiation (kJ m^{-2} day^{-1})	srad 10m	srad 5m	srad 2.5m	srad 30s
wind speed (m s^{-1})	wind 10m	wind 5m	wind 2.5m	wind 30s
water vapor pressure (kPa)	vapr 10m	vapr 5m	vapr 2.5m	vapr 30s

Future climate data

The data available here are CMIP6 downscaled future climate projections. The downscaling and calibration (bias correction) was done with WorldClim v2.1 as baseline climate.

Weather

Monthly values of minimum temperature, maximum temperature, and precipitation were processed for 23 global climate models (GCMs), and for four Shared Socio-economic Pathways (SSPs): 126, 245, 370 and 585.

The monthly values were averages over 20 year periods (2021-2040, 241-2060, 2061-2080, 2081-2100). The following spatial resolutions are available (expressed as minutes of a degree of longitude and latitude): 10 minutes, 5 minutes, 2.5 minutes, and 30 seconds.

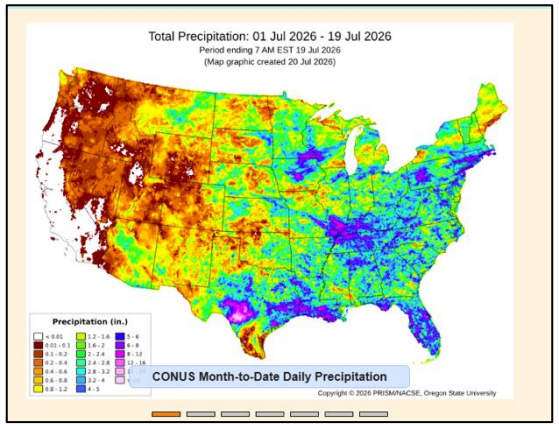
Data Layer Sources

PRISM

What?
- Precipitation, temperature, vapor pressure deficit, dewpoint, solar irradiance
  - More precise than WorldClim
  - Lots of tools to explore!

Where?
- United States only

When?
- 1895 - present



Key website sections:

Time Series Data — Download PRISM time series data grids covering daily (1981-present), monthly (1895-present), and annual (1895-present) temporal periods. Both 4km and 800m spatial resolutions available.

30-Year Normals — Download PRISM 30-year normals grids. At the end of each decade, average values for all variables are computed over the preceding 30 years. The current set of 30-year normals covers the period 1991-2020.

Data Explorer — Retrieve and download 4km and 800m time series data for point locations. Our bulk option allows up to 500 points in a single request.

Bulk Data Downloads — Access to web services and FTP resources for downloading PRISM data in bulk.

Anomalies — View "anomaly" map graphics that represent how observed values deviate from long-term conditions.

Data Directory — Web-browseable reflection of our FTP server that includes all PRISM time series and normals data, a map graphics archive, and prior versions of PRISM data.

Publications — Full listing of PRISM-related peer reviewed publications, with links to download journal articles.

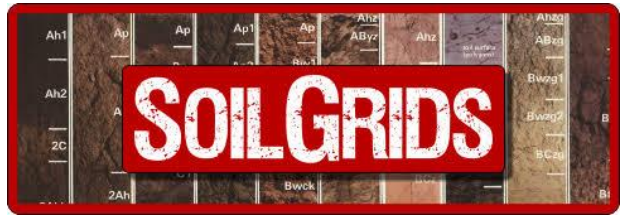
Data Layer Sources: SoilGrids

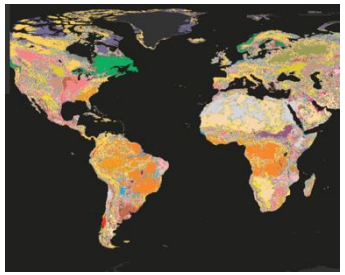
https://soilgrids.org/

What?
- 8 physical, 4 chemical, and 2 derived soil properties
  - ~ 250 m resolution
  - Include predicted values (and uncertainty layers)

Where?
- Global soils

When?
- Current day





DERIVED PROPERTIES

- ○ Organic carbon density
  in hg/m³
- ○ Soil organic carbon stock
  in t/ha

CHEMICAL SOIL PROPERTIES

- ◉ Cation exchange capacity (at ph 7)
  in mmol(c)/kg
- ○ Nitrogen
  in cg/kg
- ○ Soil organic carbon
  in dg/kg
- ○ pH water
  in pH*10

PHYSICAL SOIL PROPERTIES

- ○ Bulk density
  in cg/cm³
- ○ Clay content
  in g/kg
- ○ Coarse fragments
  in cm³/dm³
- ○ Sand
  in g/kg
- ○ Silt
  in g/kg
- ○ Vol. water content at -10 kPa
  in (10−2 cm3 cm−3)*10
- ○ Vol. water content at -33 kPa
  in (10−2 cm3 cm−3)*10
- ○ Vol. water content at -1500 kPa
  in (10−2 cm3 cm−3)*10

Name	Description	Mapped Units	Conversion Factor	Conventional Units
bdod	Bulk density	cg/cm^3	100	kg/dm^3
cec	CEC buffered at pH7	mmol(c)/kg	10	cmol(c)/kg
cfvo	Coarse fragments	cm^3/dm^3 (vol‰)	10	$cm^3/100cm^3$ (vol%)
clay	Clay	g/kg	10	g/100g (%)
nitrogen	Nitrogen	cg/kg	100	g/kg
ocd	Organic carbon density	hg/m^3	10	kg/m^3
ocs	Organic carbon stocks	t/ha	10	kg/m^2
soc	Soil organic carbon	dg/kg	10	g/kg
phh2o	pH water	pH x 10	10	-
sand	Sand	g/kg	10	g/100g (%)
silt	Silt	g/kg	10	g/100g (%)
wv0010	Volumetric Water Content at 10kPa	10^{-3} cm^3cm^{-3} (0.1 v% or 1 mm/m)	10	10−2 cm3 cm−3 (%)
wv1500	Volumetric Water Content at 1500kPa	10^{-3} cm^3cm^{-3} (0.1 v% or 1 mm/m)	10	10−2 cm3 cm−3 (%)
wv003	Volumetric Water Content at 33kPa	10^{-3} cm^3cm^{-3} (0.1 v% or 1 mm/m)	10	10−2 cm3 cm−3 (%)

Data Layer Sources: Web Soil Survey

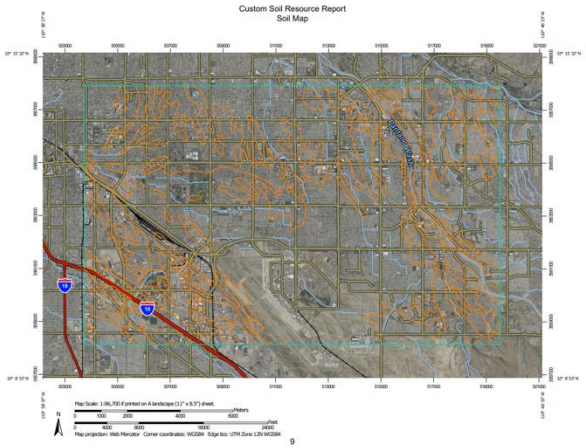
https://websoilsurvey.sc.egov.usda.gov/app/



What?
- Site-specific soil characterizations
  - Much finer resolution, but hard to get large areas
  - Need to Define Area of Interest (vectors, not a raster)

Where?
- United States only

When?
- Current day



47—Mohave soils and urban land, 1 to 8 percent slopes

Map Unit Setting
National map unit symbol: 1t0q
Elevation: 2,200 to 3,300 feet
Mean annual precipitation: 10 to 12 inches
Mean annual air temperature: 64 to 70 degrees F
Frost-free period: 220 to 280 days
Farmland classification: Not prime farmland

Map Unit Composition
Mohave and similar soils: 51 percent
Urban land: 49 percent
Estimates are based on observations, descriptions, and transects of the mapunit.

Typical profile
A - 0 to 3 inches: loam
BA - 3 to 6 inches: sandy loam
Btk - 6 to 40 inches: sandy clay loam
2C - 40 to 60 inches: loam

Properties and qualities
Slope: 1 to 8 percent
Depth to restrictive feature: More than 80 inches
Drainage class: Well drained
Runoff class: Medium
Capacity of the most limiting layer to transmit water (Ksat): Moderately high (0.20 to 0.60 in/hr)
Depth to water table: More than 80 inches
Frequency of flooding: None

Data Layer Sources: EPA

https://www.epa.gov/enviroatlas

What?
- Air quality, bodies of water, land use, boundaries, housing, and much more
  - Interactive tool and downloadable data
  - Selected high-population regions at higher resolution

Where?
- United States only

When?
- Current day
  - some layers may have historical data

Boundaries

- Ecological Boundaries
- Hydrologic Features
- Political Boundaries

People and Built Spaces

- Commuting and Walkability
- Employment
- Housing and Schools
- Population Distribution
- Quality of Life
- Vacancy

Pollution Sources and Impacts

- Harmful Algal Blooms
- Impaired Waters
- National Air Toxics Assessment
- Pollutants: Nutrients
- Pollutants: Other
- Sites Reporting to EPA

Ecosystem Services and Biodiversity

- Carbon Storage
- Crop Productivity
- Ecosystem Markets
- Energy Potential
- Engagement with Outdoors
- Health and Economic Outcomes
- Land Cover: Near-Water
- Land Cover: Type
- Landscape Pattern
- Livestock and Poultry Production
- Near-Road Environments
- Pollutant Reduction: Air
- Pollutant Reduction: Water
- Protected Lands
- Soils
- Species: At-Risk and Priority
- Species: Other
- Water Supply, Runoff, and Flow
- Water Use
- Weather and Climate
- Wetlands and Lowlands

Other Data Layer Sources

Paleoclimate data

- NOAA/NCEI: ncei.noaa.gov/products/paleoclimatology
- PMIP: pmip.lsce.ipsl.fr
- PaleoClim: paleoclim.org

Aquatic Environments

- AquaMaps: aquamaps.org/main/envt_data.php

Hourly or Daily data

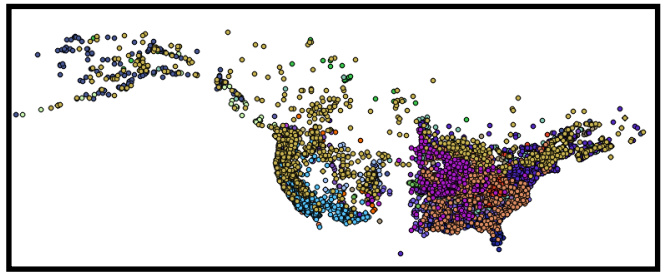
- Daymet: daymet.ornl.gov
- OGIMET: ogimet.com

Data Exploration

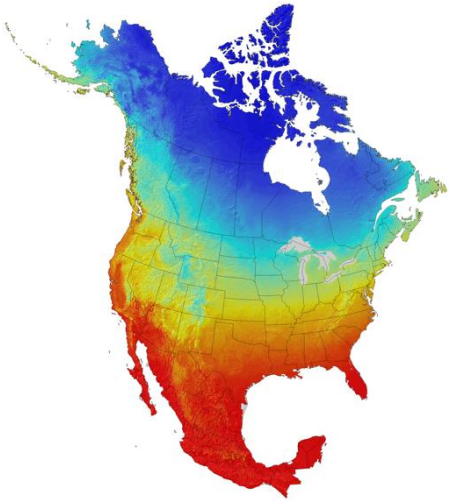
Goal:

Examine the data to determine variables of interest, possible issues, and developing trends

Using:

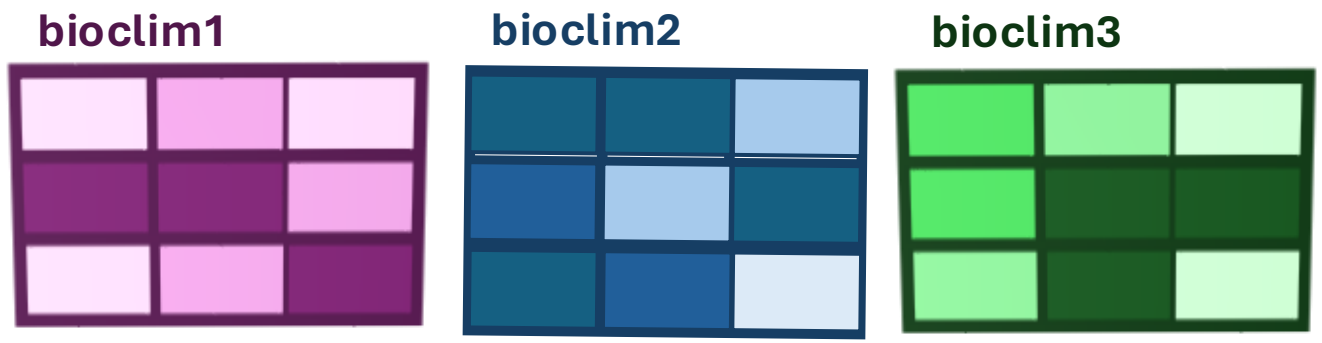


Cleaned, georeferenced occurrences

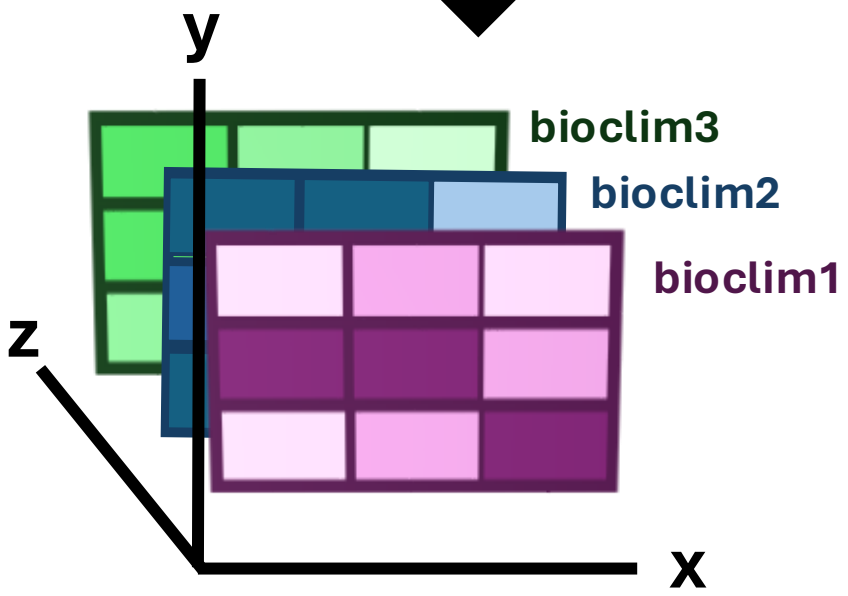


Environmental variables of interest (rasters)

Extracting Climatic Variables at Occurrences

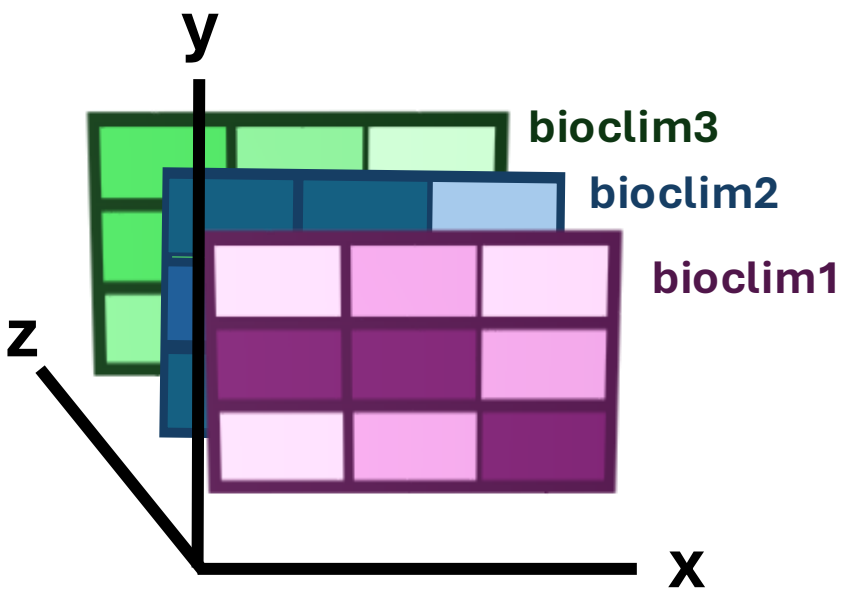


Stack the rasters

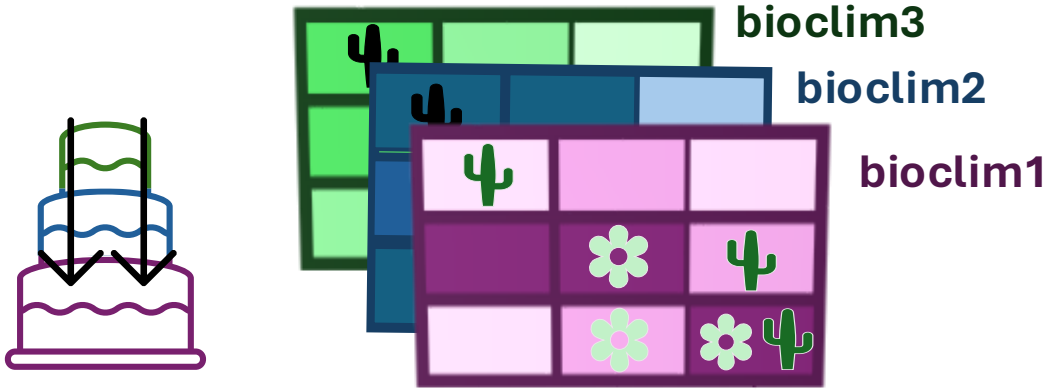




Extracting Climatic Variables at Occurrences

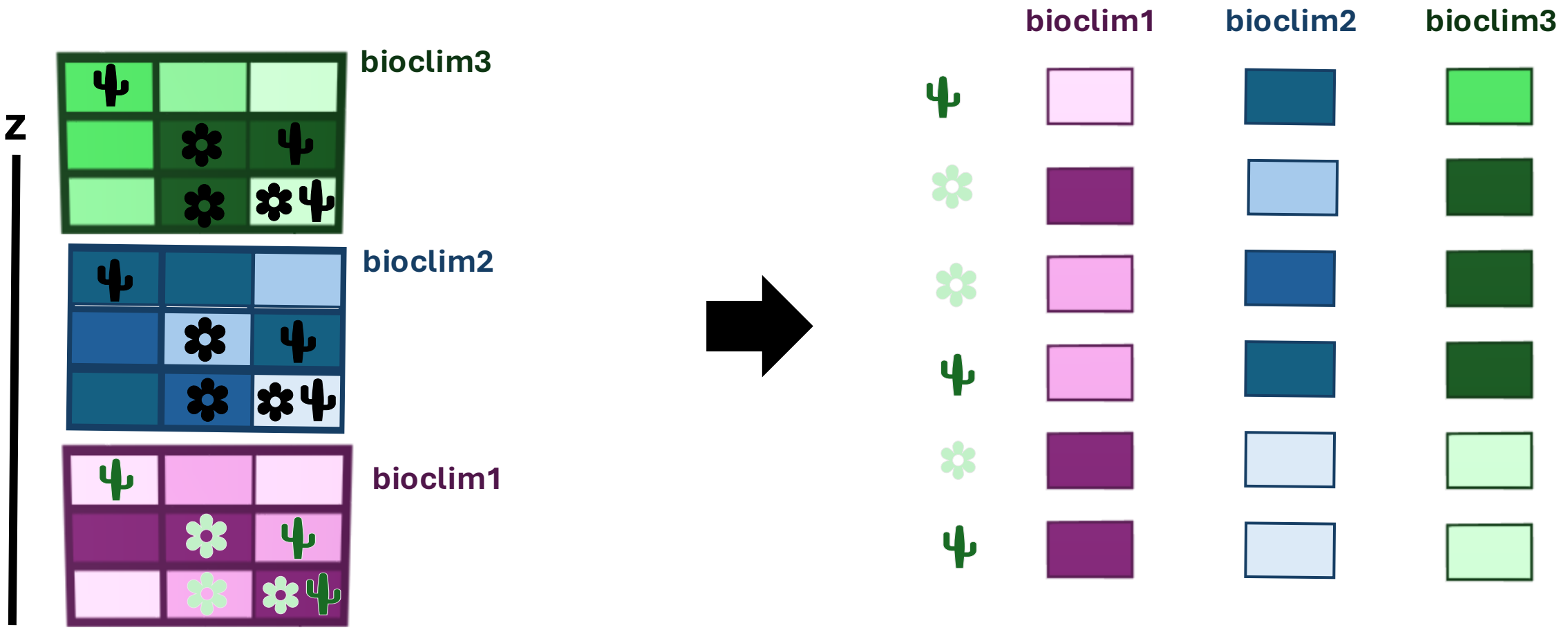


Extracting Climatic Variables at Occurrences



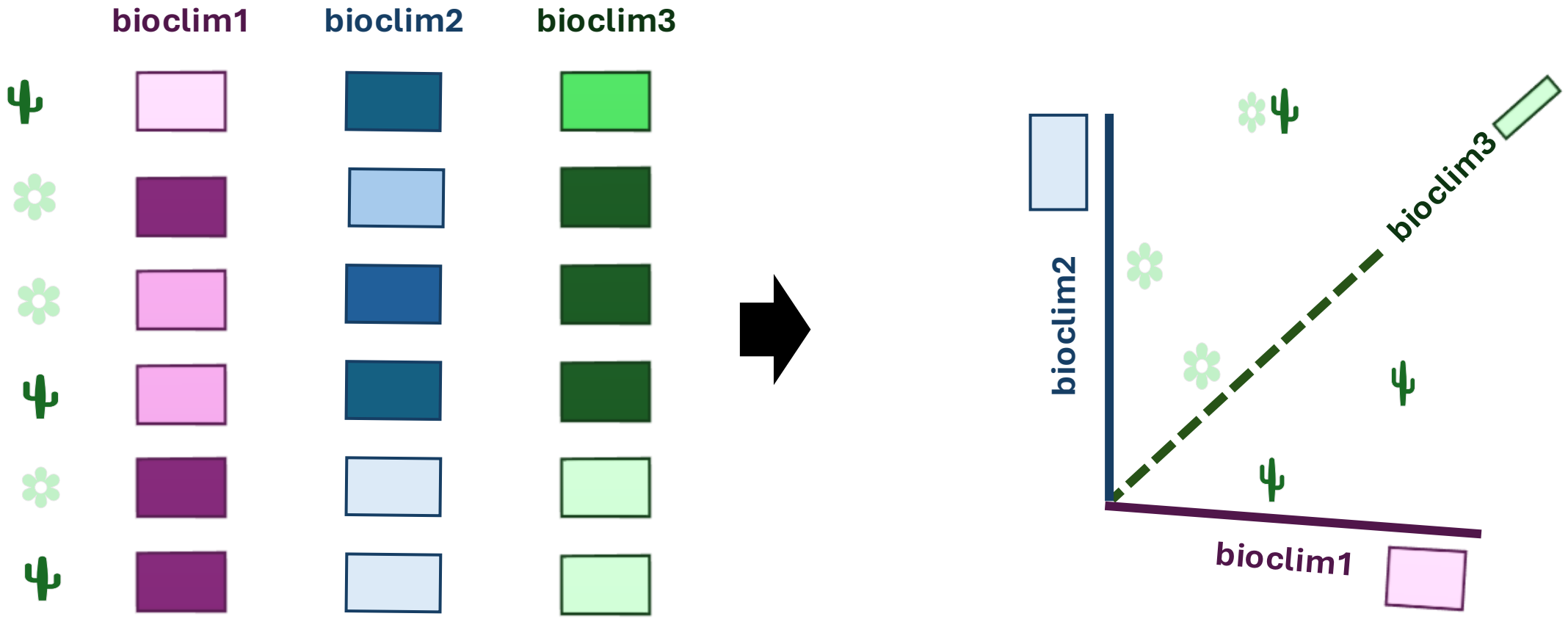
Extract data from raster for each occurrence point

Extracting Climatic Variables at Occurrences

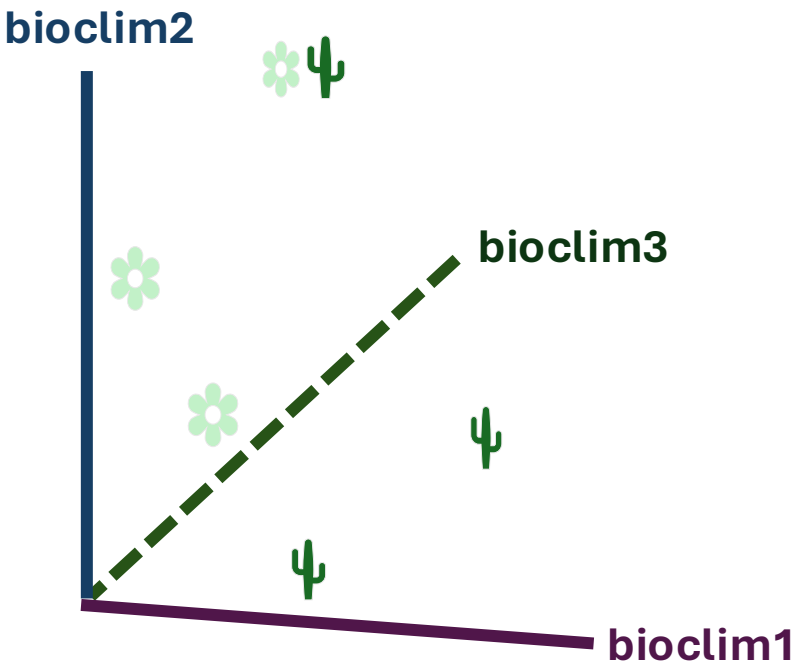


Extract data from raster for each occurrence point

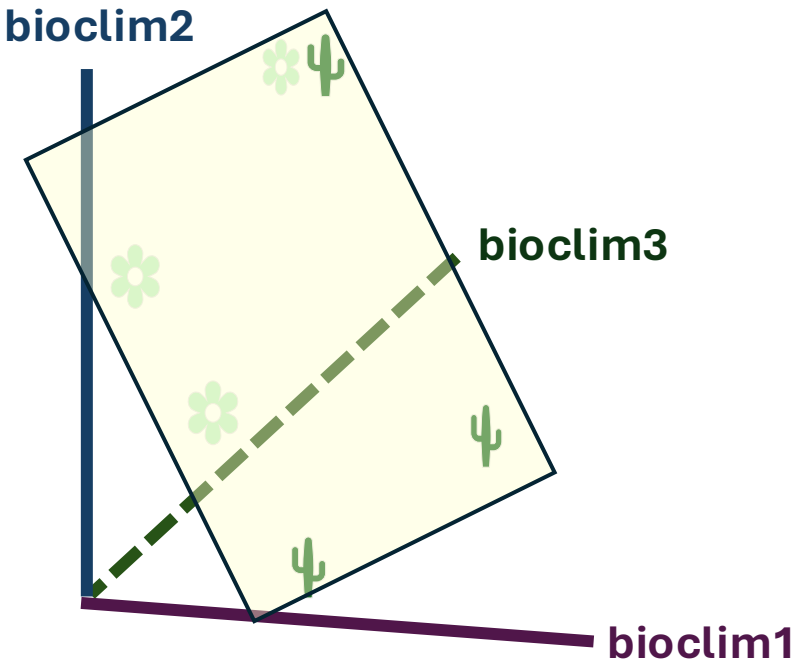
Principal Component Analysis



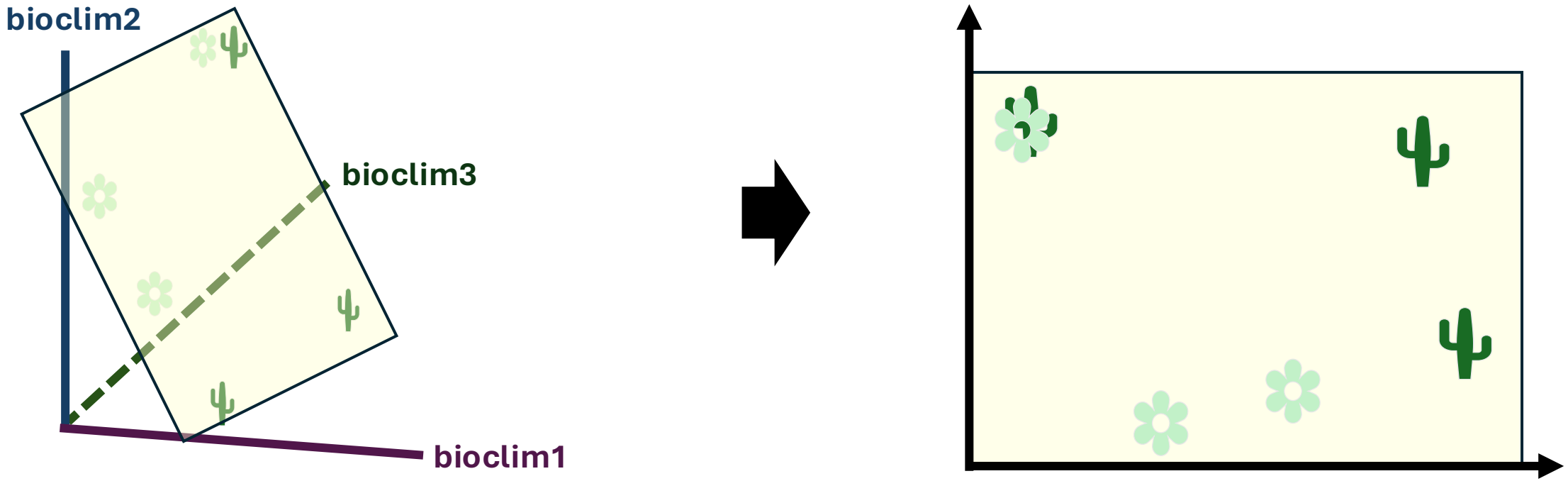
Principal Component Analysis



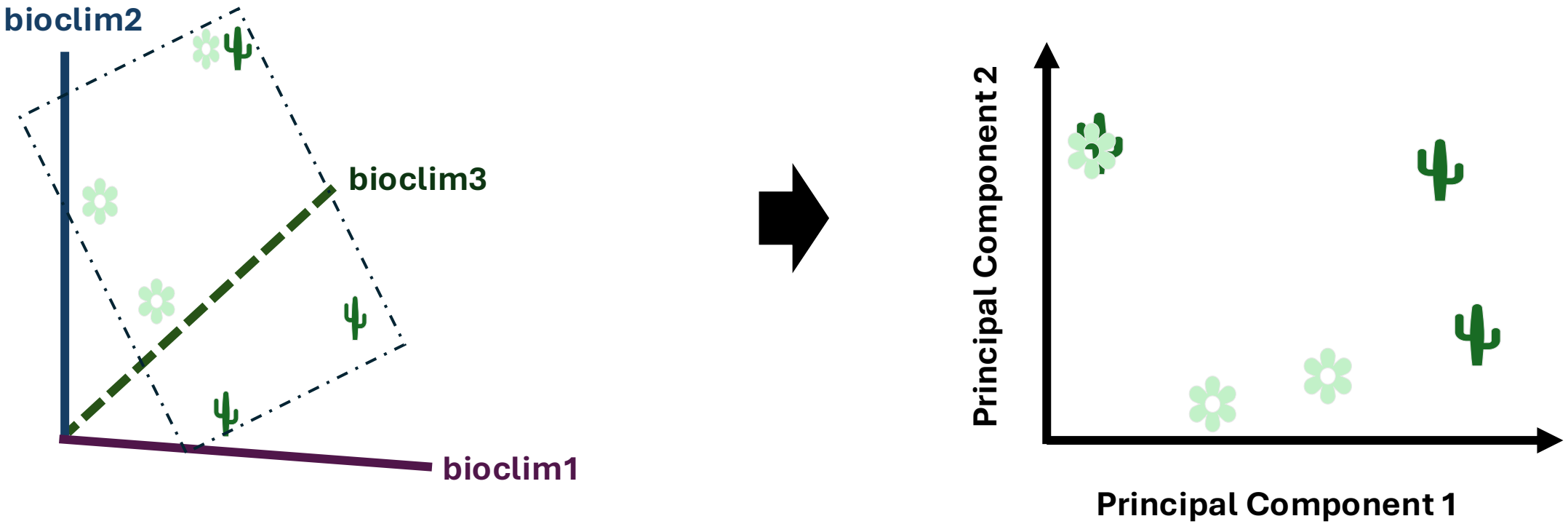
Principal Component Analysis



Principal Component Analysis

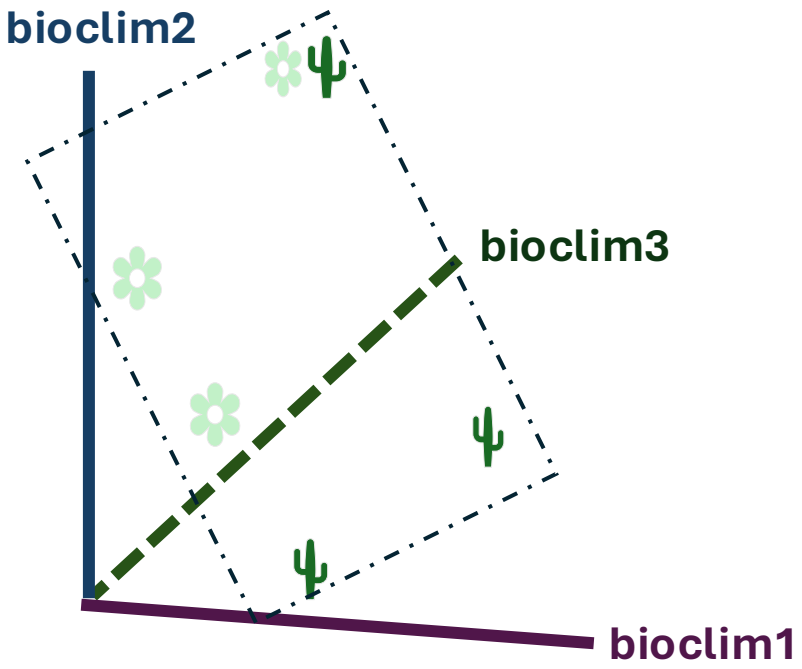


Principal Component Analysis



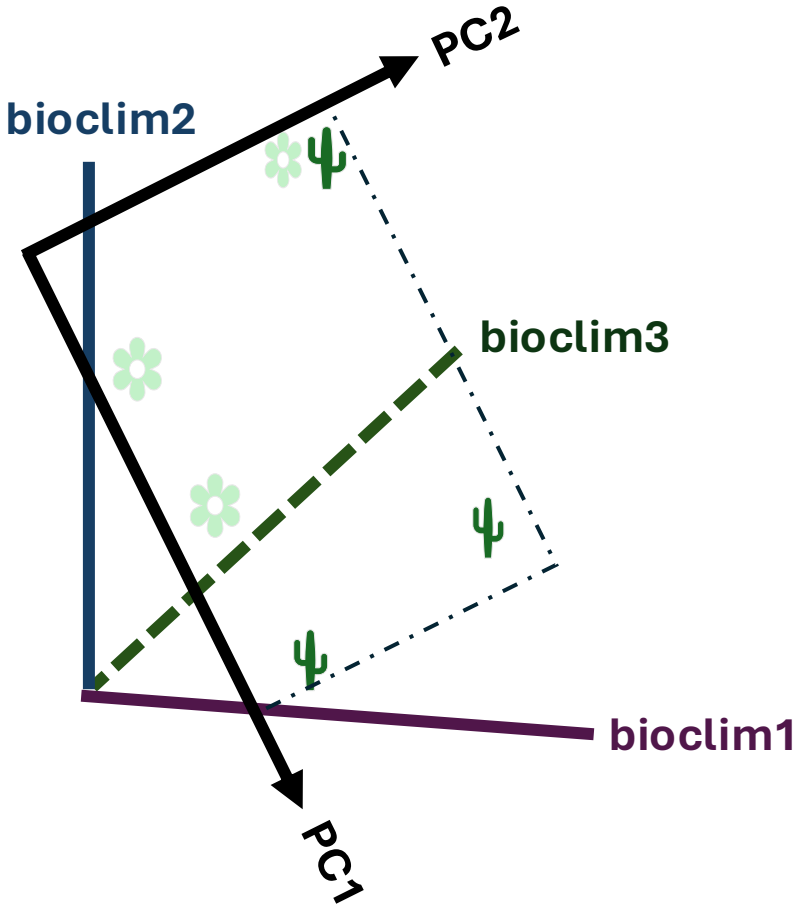
Transforms data into new set of variables, called **Principal Components (PCs)**

Principal Component Analysis



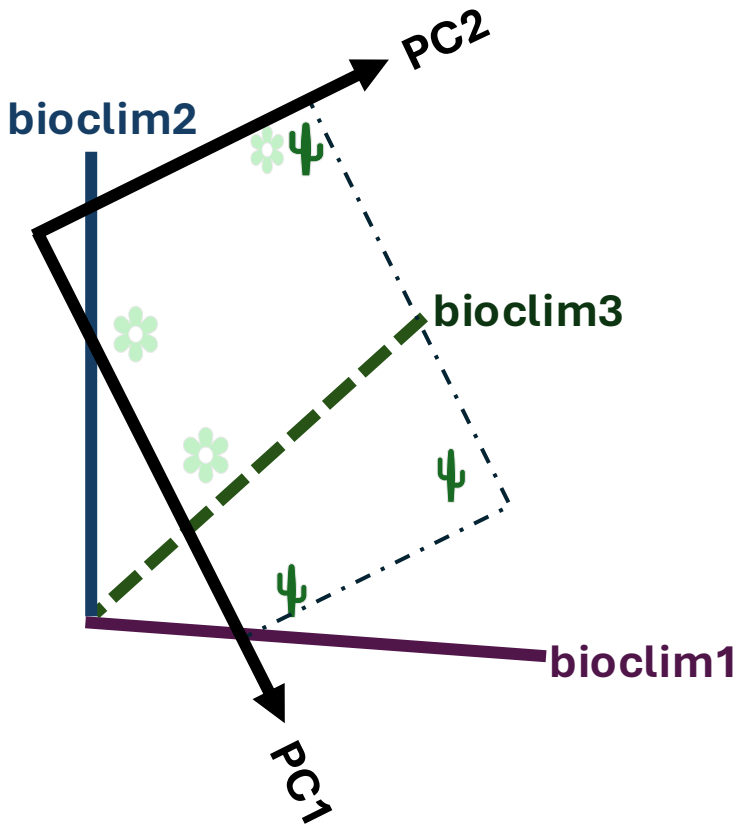
Loading: How much of each original variable contributes to a Principal Component

Principal Component Analysis



Loading: How much of each original variable contributes to a Principal Component (PC)

Principal Component Analysis



PC1:

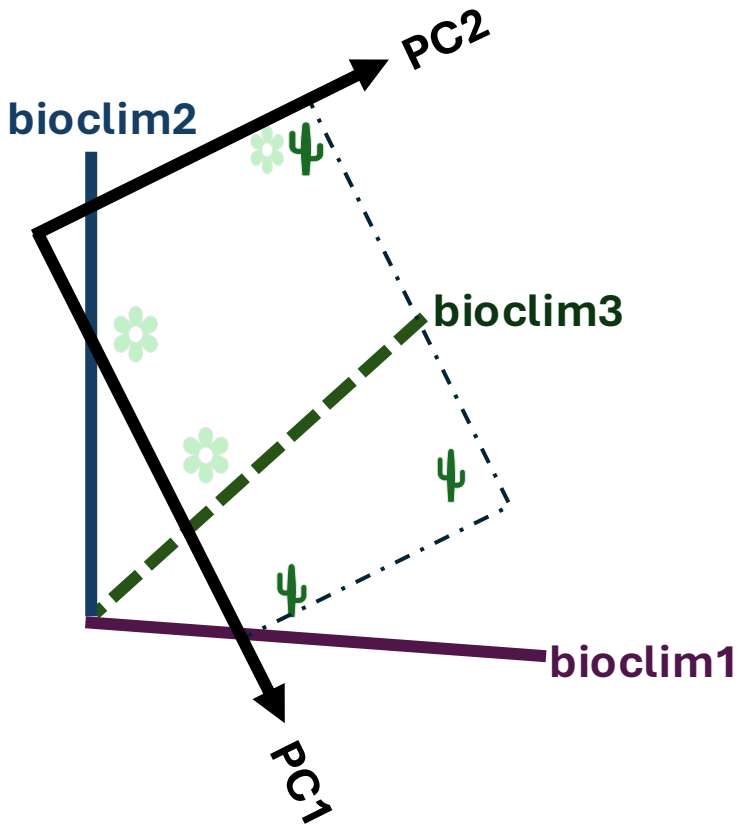
Variable	*Loading*
Bioclim2	-1.45636
Bioclim1	*0.45256*
Bioclim3	*0.11324*

PC2:

Variable	*Loading*
Bioclim3	2.65567
Bioclim1	*2.34566*

Loading: How much of each original variable contributes to a Principal Component (PC)

Principal Component Analysis



PC1:

Variable	*Loading*	*Relative Loading*
Bioclim2	-1.45636	72.0%
Bioclim1	*0.45256*	*22.4%*
Bioclim3	*0.11324*	*5.6%*
	2.0221	

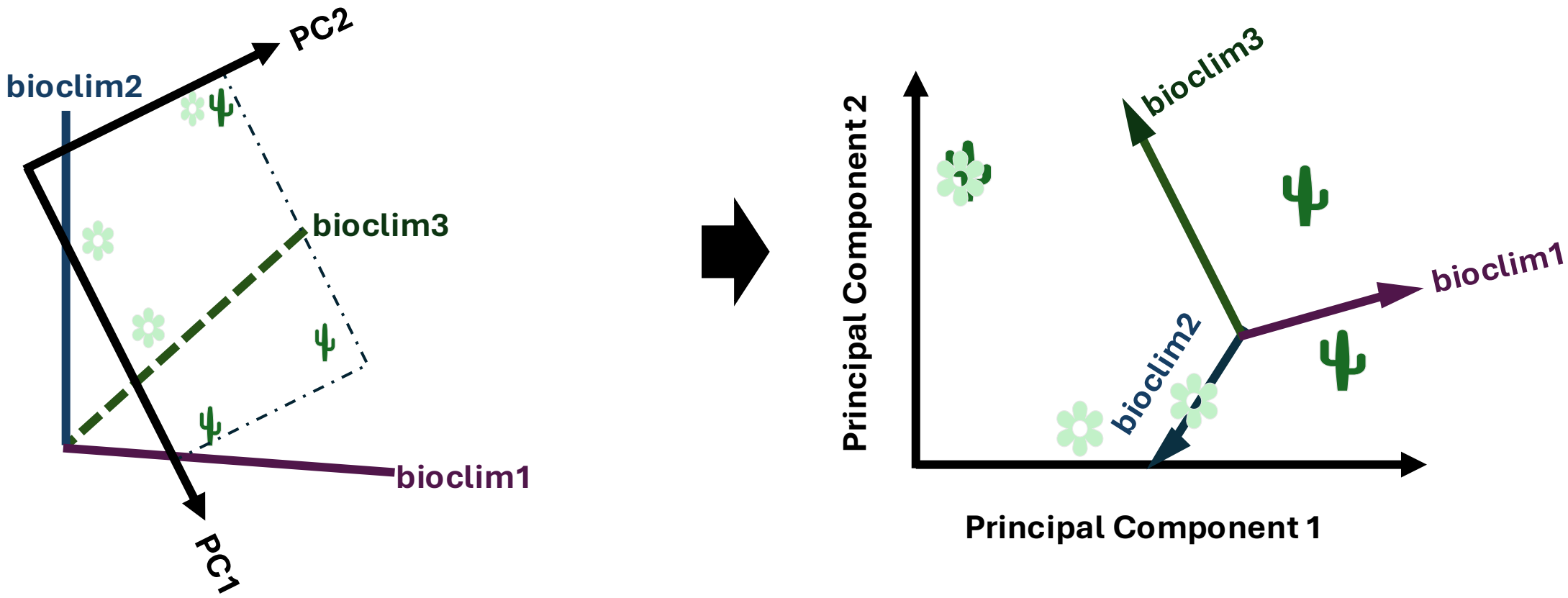
PC2:

Variable	*Loading*	*Relative Loading*
Bioclim3	2.65567	53.1%
Bioclim1	*2.34566*	*46.9%*
	5.00133	

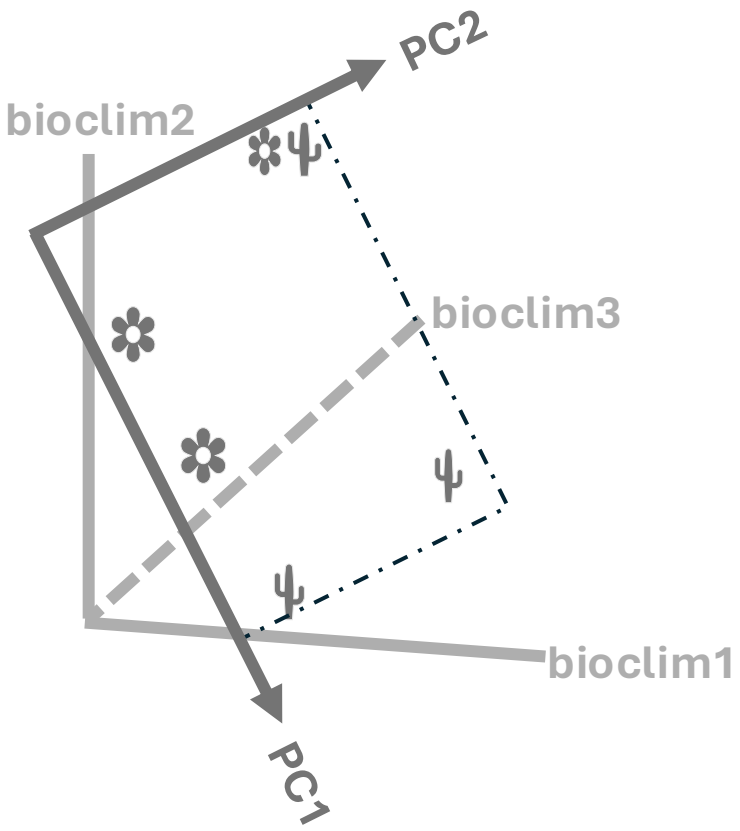
Loading: How much of each original variable contributes to a Principal Component (PC)

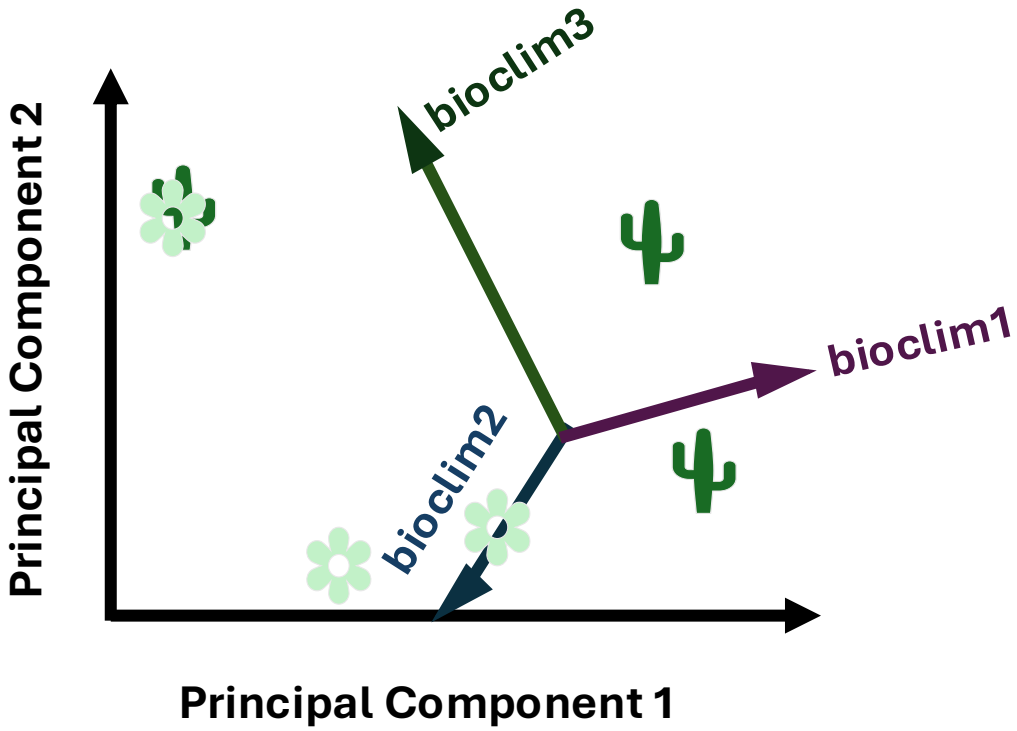
Can calculate and examine the relative contribution of each climatic variable to each PC

Principal Component Analysis

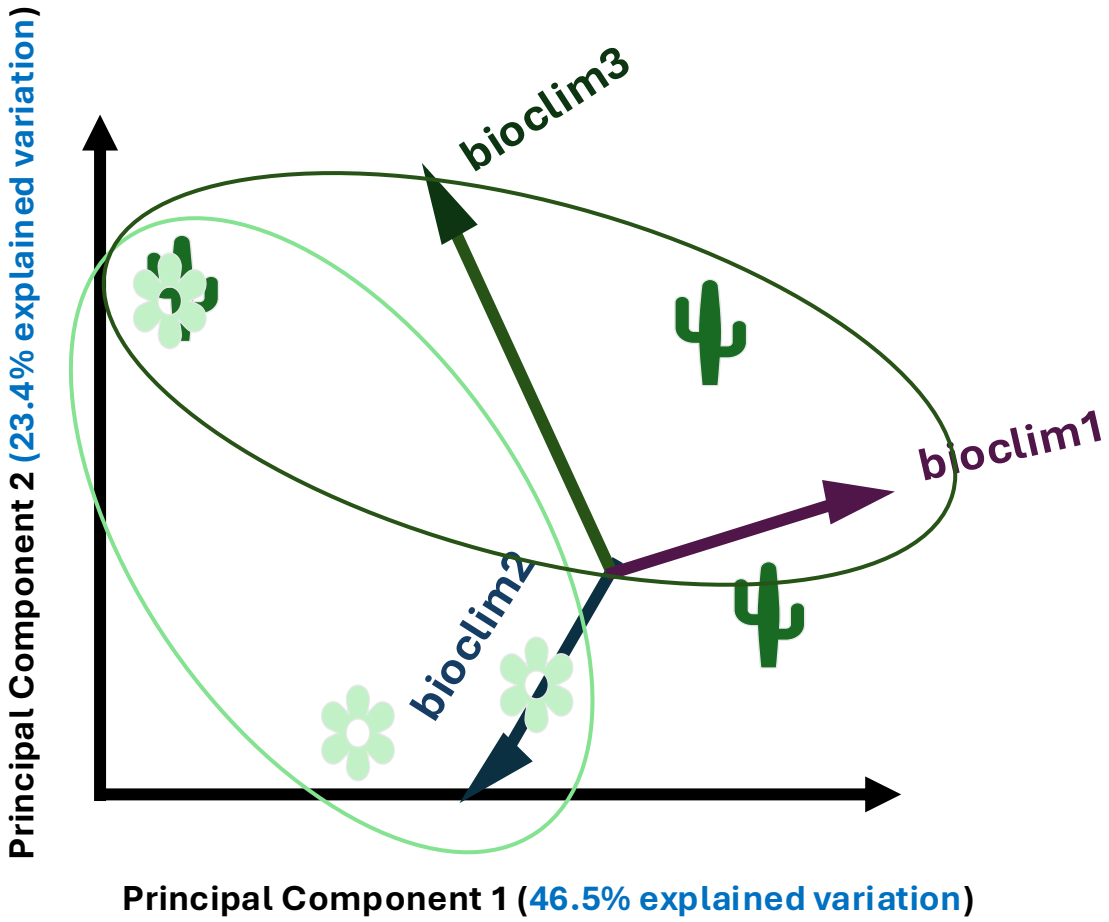


Principal Component Analysis

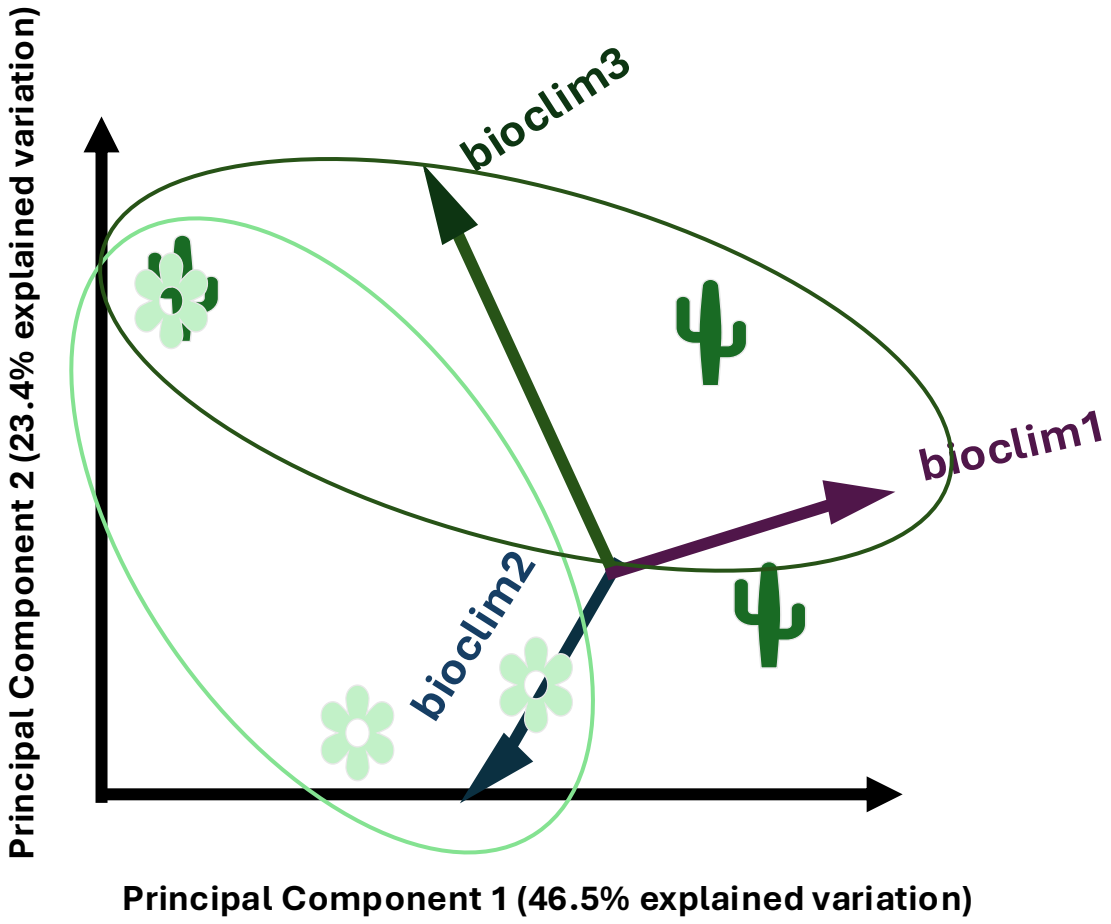




Principal Component Analysis



Principal Component Analysis (PCA)



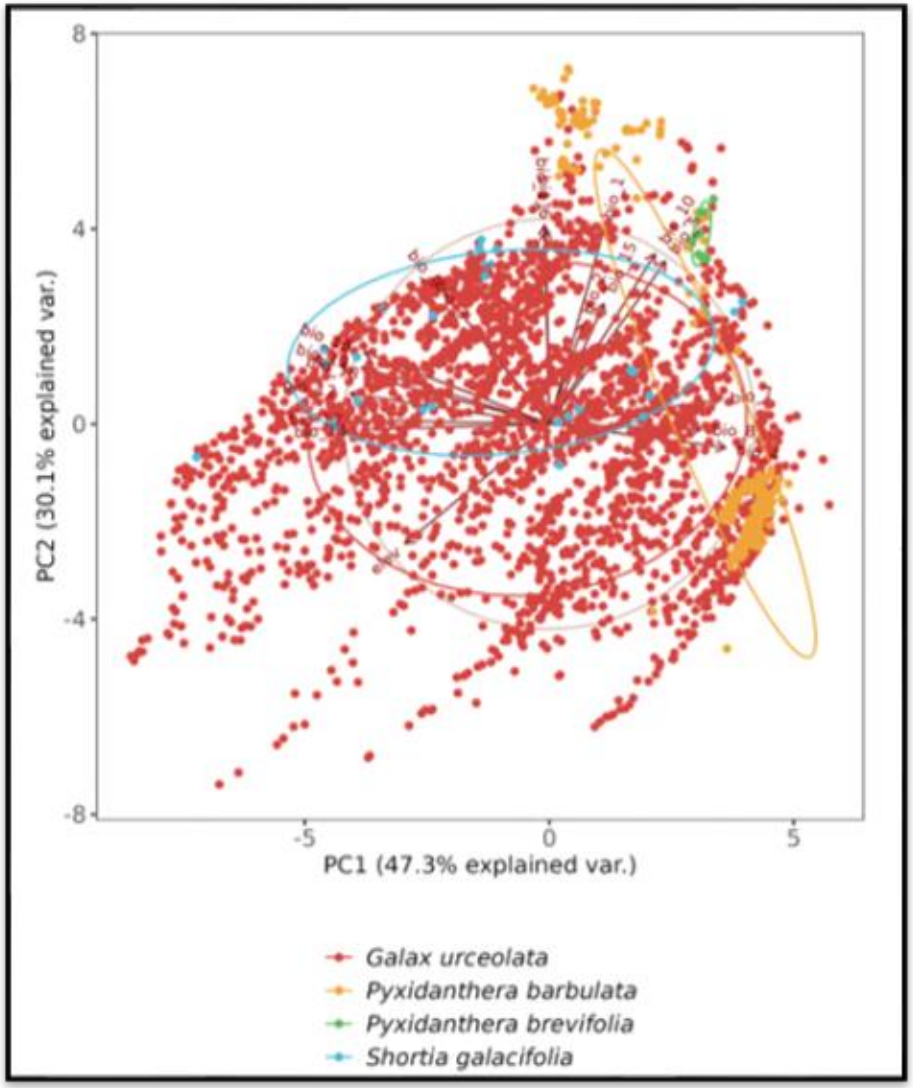
Reading PCAs:

- *Points: observations*
- Arrows: environmental variables
- *Ellipse groupings:* taxon

PCAs:

- *Summarize* many environmental variables into PCs
- *Reveal groupings* that are not obvious in the raw data

Principal Component Analysis (PCA)



Reading PCAs:

- *Points: observations*
- Arrows: environmental variables
- *Ellipse groupings:* taxon

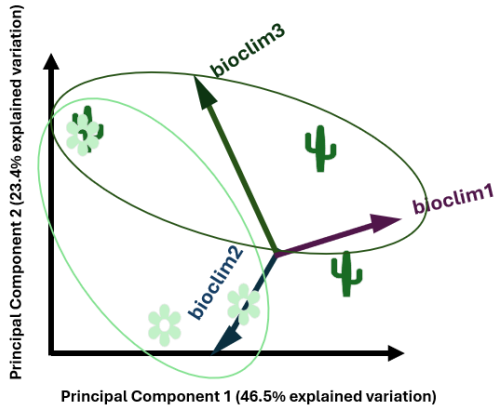
PCAs:

- *Summarize* many environmental variables into PCs
- *Reveal groupings* that are not obvious in the raw data

ANOVA

ANOVA Test:

- Select the **top two loading variables** for the **top two principal components**



PC1:

46.5% of explained variation

PC2:

23.4% of explained variation

PC3:

8.4% of explained variation

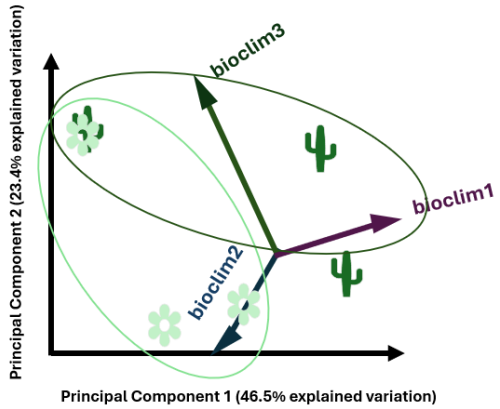
PC4:

5.7% of explained variation

ANOVA

ANOVA Test:

- Select the **top two loading variables** for the **top two principal components**



PC1:

46.5% of explained variation

PC2:

23.4% of explained variation

PC3:

8.4% of explained variation

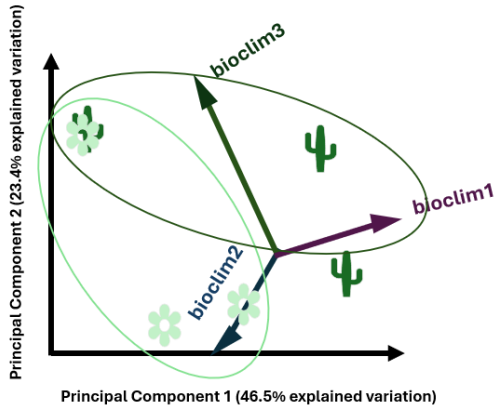
PC4:

5.7% of explained variation

ANOVA

ANOVA Test:

- Select the **top two loading variables** for the **top two principal components**



PC1:

46.5% of explained variation

Variable	Loading	Relative Loading
Bioclim2	-1.45636	72.0%
Bioclim1	*0.45256*	*22.4%*
Bioclim3	*0.11324*	*5.6%*

PC2:

23.4% of explained variation

Variable	Loading	Relative Loading
Bioclim3	2.65567	53.1%
Bioclim1	*2.34566*	*46.9%*

Bioclim 2

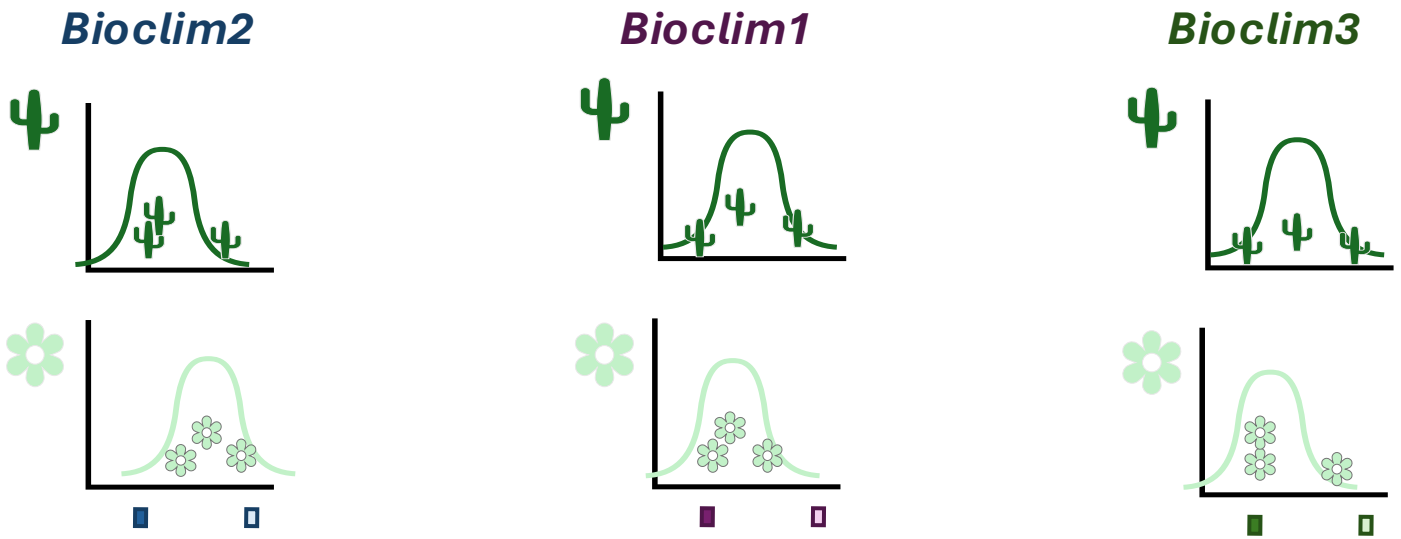
Bioclim 1

Bioclim 3

ANOVA

ANOVA Test:

- Select the **top two loading variables** for the **top two principal components**
- For each of these **top loading variables**, perform an ANOVA test to see if taxa are significantly different



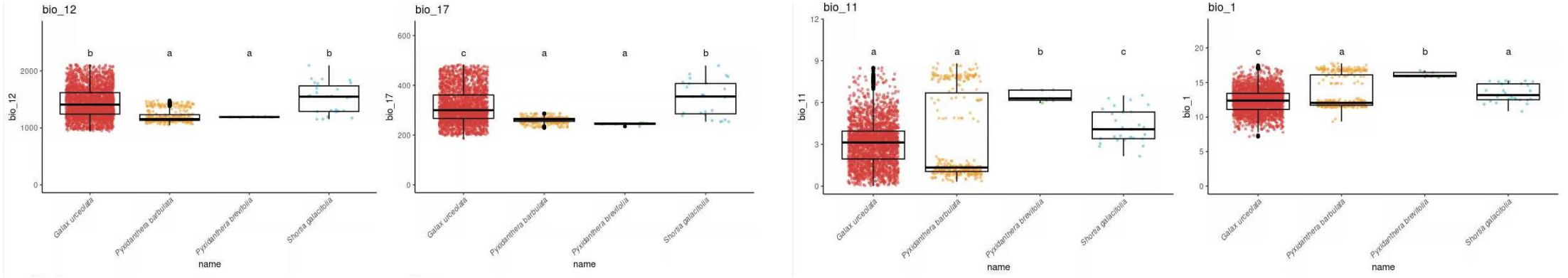
ANOVA

ANOVA Test:

- Select the **top two loading variables** for the **top two principal components**
- For each of these **top loading variables**, perform an ANOVA test to see if taxa are significantly different

Tukey HSD test:

- Determines **which taxa** are significantly similar/different

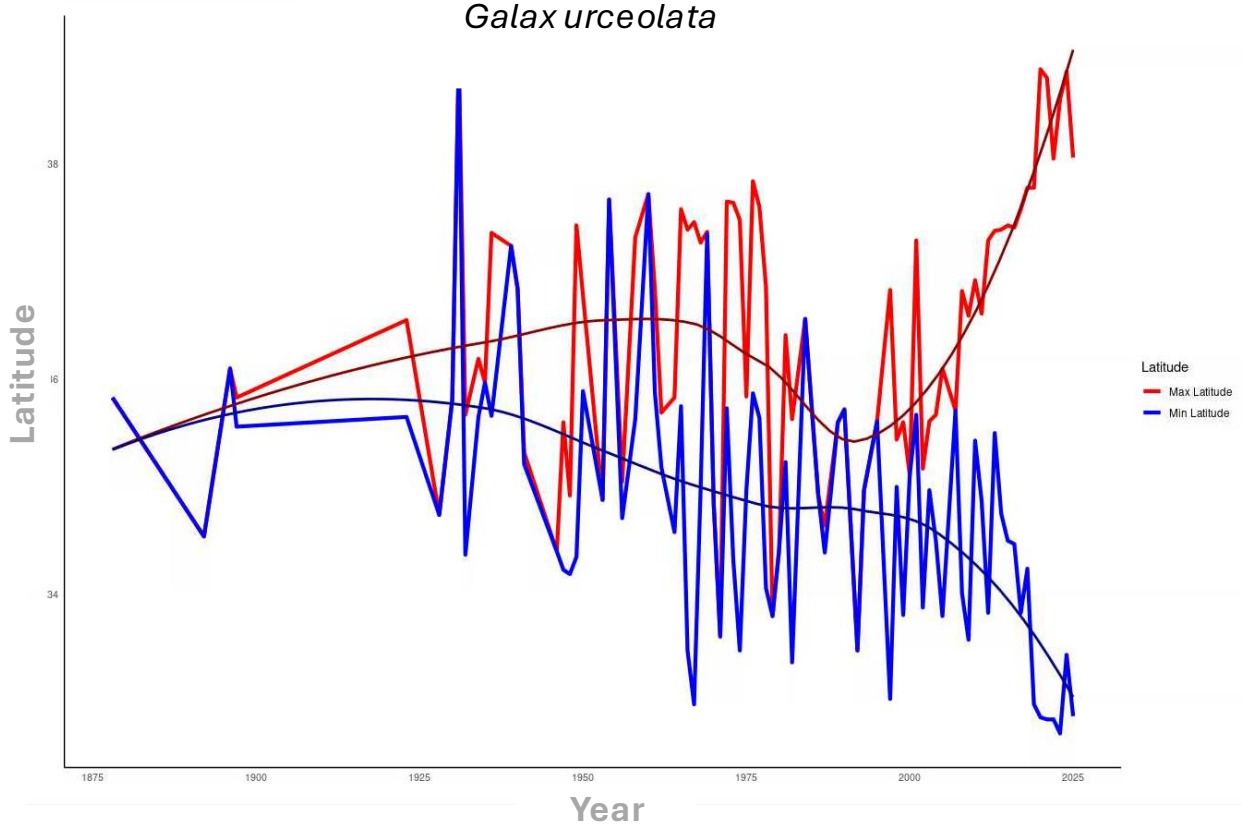


Caution!

Do not just rely on environmental variables

- Uses dates of occurrence records to identify trends in distribution over time
- Consider potential impacts of increases in volume of occurrence records over time

Minimum and Maximum Latitude Over Time



Importance

Data Exploration can reveal:

- ! Potential issues, biases, and discrepancies in raw data
-  Trends that are worth exploring further
-  Population/species differences

Data exploration allows us to examine the reliability of data before putting it into the ENM models



Coding Practical

03_Data_Exploration.R